

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 91-155

WASTE DISCHARGE REQUIREMENTS
FOR
MOUNT VERNON COUNTY SANITATION DISTRICT
WASTEWATER TREATMENT FACILITY
KERN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Mount Vernon County Sanitation District (hereafter Discharger) submitted a Report of Waste Discharge, dated 2 July 1990, and a technical report, dated June 1990, to improve the effluent quality of the existing wastewater treatment facility (WWTF). The WWTF and the disposal area (about 1400 acres) properties are owned by the Discharger (Assessor's Parcel Nos. 173-191-02, 173-191-03, 173-191-06, 173-191-07, 173-220-03, 173-220-04, 173-220-05, 173-220-06, 173-220-09, 173-220-10, 173-220-11, 173-220-12, 173-220-13, 173-220-14, 174-100-01, 174-100-02, 174-180-10, and 174-180-11).
2. Waste Discharge Requirements Order No. 72-282, adopted by the Board on 23 June 1972, prescribes requirements for discharge of 6.6 million gallons per day (mgd). The existing plant consists of two treatment trains; one built in 1948 to provide secondary treatment of 3.0 million gallons per day (mgd); the other added in 1966 to provide for primary treatment of an additional 3.6 mgd. However, present flows average 4.0 mgd, and influent BOD and suspended solids daily loadings have nearly doubled since 1972. Consequently, the Discharger has been consistently in noncompliance with the 40 mg/l effluent limitations for BOD₅ and suspended solids established by WDRs Order No. 72-282. Attachment A, which is hereto made part of this Order by reference, shows the existing WWTF.
3. Influent consists of domestic and industrial wastes. The industrial sources include three electroplating shops, Kern Medical Center, the Kern County Health Department, and the Kern County Coroner, all of which are subject to categorical pretreatment standards; a company which produces oilfield related chemicals, and several auto repair shops, restaurants, commercial establishments, and chemical laboratories. Based on a 1990-1991 survey of industrial dischargers, the industrial flow is about 8 percent (298,000 gallons per day) of the average flow into the plant. Considering these facts, a pretreatment program is appropriate.
4. The Board adopted a Water Quality Control Plan for the Tulare Lake Basin (5D) (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
5. The Basin Plan states that the minimum treatment level for WWTFs which discharge more than 1.0 mgd is secondary treatment, which is defined in

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the Basin Plan as 80 percent removal of BOD₅ and suspended solids, or reduction to 40 mg/l, whichever is more restrictive.

6. The Discharger proposes to discharge 7.0 mgd of secondary treated wastewater by adding unit processes to the primary treatment train so that it provides secondary treatment for about 3.0 mgd. The proposed improvements include a trickling filter and recirculation pump, two secondary clarifiers, sludge pumping facilities, an effluent structure, and associated pumping facilities, and are illustrated in Attachment B.
7. Effluent from the existing WWTF is pumped to a storage reservoir of unspecified capacity about one-half mile south of the plant and reclaimed by surface irrigation on about 1400 acres of cotton, fiber, and seed crops on land (hereafter referred to as designated reclamation area) adjacent to the WWTF and owned by the Discharger. The Discharger proposes to continue this method of wastewater reclamation for the proposed 7.0-mgd discharge.
8. The proposed discharge for reclamation purposes is subject to reclamation criteria contained in Chapter 3, Division 4, Title 22, California Code of Regulations (CCR), hereafter Title 22. This Order prescribes reclamation requirements consistent with Title 22.
9. For the past 16 years, sludge from the existing plant has been disposed of by landspreading on a 20-acre parcel adjacent to the WWTF and owned by the Discharger (Assessor's Parcel No. 173-191-06). Soils in the sludge and effluent disposal areas were tested and found to have relatively high concentrations of metals, which are of sufficient concern that soils and ground water monitoring are appropriate.
10. The WWTF is in Section 3, T30S, R28E, MDB&M, and the reclamation areas in Sections 3, 10, 15, and 27, T30S, R28E, MDB&M, with surface water drainage to the Valley floor, as shown in Attachment A. The site lies within the Kern Delta Hydrologic Area (No. 557.10), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
11. Ground water in the area is 140 to 190 below the ground surface (bgs), of excellent quality (EC of about 250 umhos/cm, pH of 7.6, total dissolved solids of 160 mg/l), with a northeasterly gradient.
12. The Basin Plan identifies beneficial uses of ground water throughout the Tulare Lake Basin as municipal and domestic supply, industrial service supply, industrial process supply, water contact recreation, and wildlife habitat.
13. Soils below the effluent disposal area consist of sands, silty sands, silty clays, and clays with moderate to low permeabilities.

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14. The action to revise waste discharge requirements for the existing facility is exempt from the provisions of the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.), in accordance with Section 15301, Title 14, California Code of Regulations.
15. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
16. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 72-282 is rescinded and Mount Vernon County Sanitation District, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at the above-described facility:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, Title 23, CCR (hereafter Chapter 15), is prohibited.
4. The use of untreated or partially treated waste for irrigation is prohibited.

B. Discharge Specifications:

1. The monthly average daily wet weather discharge flow shall not exceed 7.0 million gallons.
2. The monthly average dry weather discharge flow shall not exceed 6.6 million gallons.
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.

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4. As means of discerning compliance with Discharge Specification No. B.3, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds or any impoundment shall not be less than 1.0 mg/l.
5. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
6. The effluent from the treatment facility shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ^{1/}	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5
Total Suspended Solids	mg/l	40	80

^{1/} Five-day, 20° Celsius biochemical oxygen demand.

7. The maximum specific electrical conductivity (EC) of the discharge shall not exceed the average EC of the source water plus 500 umhos/cm, or a maximum of 900 umhos/cm, whichever is less.
8. The discharge shall not have a pH less than 6.5.
9. Use of reclaimed water shall be limited to surface irrigation of fodder, fiber, and seed crops for nonhuman consumption, and for construction purposes.
10. Irrigation or impoundment of wastewater shall not occur within 500 feet of any domestic well or within 100 feet of any irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that a lesser distance is justified.
11. There shall be no standing water in any subarea of the reclamation area 24 hours after each irrigation of that subarea ceases.
12. Storage ponds and areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitos. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

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- b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
- 13. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
 - 14. The storage reservoir shall have sufficient capacity to accommodate wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than 2 feet (measured vertically).
 - 15. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.14.
 - 16. Storm water runoff from the irrigation fields shall not be discharged to any surface water drainage course within 30 days of the last wastewater application.

C. Sludge Disposal:

- 1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
- 2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 60 days in advance of the change.
- 3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, and with Clean Water Act (CWA) Section 405(d) technical standards when promulgated.

If an applicable management practice or numerical limitation for pollutants in sewage sludge is promulgated under Section 405(d) of the CWA after issuance of this permit and is more stringent than the sludge pollutant limit or management practice specified in this permit or in existing federal or state laws or regulations, this order shall be promptly modified or revoked and reissued to conform to the regulations promulgated under Section 405(d) of the CWA. The Discharger shall comply with the limitations by no later than the

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compliance deadline specified in the applicable regulations as required by Section 405(d) of the CWA.

D. GROUND WATER LIMITATIONS

The Discharge, in combination with other sources, shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified. (For purposes of comparison, background water quality shall be determined when background monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards.")
2. Exceed an annual average incremental increase in specific electrical conductivity greater than 3 umhos/cm, based on the most recent five-year period, or a maximum of 900 umhos/cm., whichever is less.
3. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in Chapter 15, Division 4, Title 22, CCR.
4. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
5. Exceed concentrations of radionuclides specified in Chapter 15, Division 4, Title 22, CCR.
6. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
7. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

E. Provisions:

1. The Discharger shall comply with the attached Monitoring and Reporting Program No. 91-155, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, attached hereto and a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

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3. The Discharger shall submit technical reports as directed by the Executive Officer.
4. Reclaimed wastewater shall meet the criteria contained in Title 22 and, if used in construction, the most current edition of *"Guidelines for Use of Reclaimed Water for Construction Purposes"*, published by the State Department of Health Services. Other uses for reclaimed wastewater shall be subject to the approval of the Executive Officer.
5. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this order.
6. The Discharger shall submit to the Board an irrigation management plan for the use of reclaimed water, pursuant to the time schedule in Provision E.11.i. The irrigation management plan shall address current and design flows, describe the acreage of various types of crops to be grown and harvested annually, and discuss crop water use and nitrogen uptake data. The plan must include a water balance for current and design flows, with storage requirements, and a nitrogen balance for current and design flows, and must demonstrate that reclamation can be accomplished in accordance with accepted irrigation practices without contributing additional nitrogen in the form of nitrate ion to the ground water. The plan shall also address compliance with Discharge Specifications B.10 to B.14.
7. **Effective 1 March 1992**, the Discharger shall comply with all pretreatment requirements contained in 40 CFR Part 403 and shall be subject to enforcement actions, penalties, fines, and other remedies by EPA or other appropriate parties, as provided in the Clean Water Act, as amended. The Discharger shall implement and enforce its approved Publicly-Owned Treatment Works (POTW) Pretreatment Program, which is hereby made an enforceable condition of this Order. EPA may initiate enforcement action against an industrial user for noncompliance with applicable standards and requirements as provided in the Clean Water Act.
8. **Effective 1 March 1992**, the Discharger shall enforce the requirements promulgated under Sections 307(b), (c), (d), and 402(b) of the Clean Water Act. The Discharger shall cause industrial users subject to federal categorical standards to achieve compliance by **1 July 1993** or, in the case of a new industrial user, upon commencement of the discharge.
9. **Effective 1 March 1992**, the Discharger shall perform the pretreatment functions required in 40 CFR Part 403, including, but not limited to:

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- a. Implementing the necessary legal authorities as provided in 40 CFR 403.8(f)(1);
 - b. Enforcing the pretreatment requirements under 40 CFR 403.5 and 403.6;
 - c. Implementing the programmatic functions as provided in 40 CFR 403.8(f)(2);
 - d. Providing the requisite funding and personnel to implement the Pretreatment program as provided in 40 CFR 403.8(f)(3); and,
 - e. Publishing a list of significant violators as required by 40 CFR 403.8(f)(2)(vii).
10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
 11. The Discharger shall comply with the following time schedule to assure compliance with the specifications and provisions of this Order:

Task

Compliance Date

Pretreatment Program

- | | |
|--|-------------|
| a. Submit the final results of the industrial user survey | 15 Sep 1991 |
| b. Submit an evaluation of the legal authority necessary for the administration and enforcement of the requirements of Section 307(b) and (c) and 402(b)(8) of the Clean Water Act | 15 Aug 1991 |
| c. Submit a determination of the technical information necessary to develop and implement the pretreatment ordinance or other means of enforcing pretreatment standards | 15 Aug 1991 |
| d. Submit an evaluation of the financial programs and revenue sources to implement the program, including proposed funding and staffing levels | 15 Oct 1991 |

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<u>Task</u>	<u>Compliance Date</u>
Pretreatment Program (contd.)	
e. Submit the design of a monitoring program which will implement the requirements of the pretreatment program	15 Oct 1991
f. Submit a list of monitoring equipment required to implement the pretreatment program and a description of municipal facilities necessary for monitoring and analysis of industrial wastes	15 Dec 1991
g. Submit the effluent limitations for general prohibition pollutants (40 CFR 403.5) which shall be incorporated into the pretreatment ordinance	15 Feb 1992
h. Submit complete pretreatment program package (40 CFR 403.8) with request for pretreatment program approval	15 Feb 1992
Irrigation Management Plan	
i. Submit proposed irrigation management plan concerning compliance with Provision E.6 and report on measures to preclude public access	15 Sep 1991
j. Submit final irrigation management plan and implement measures to preclude public access	15 Nov 1991
k. Implement irrigation management plan	15 Apr 1992
12. The Discharger shall submit to the Board on or before each report due date, a report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board by letter when compliance with the time schedule resumes.	
13. The Discharger shall provide a letter report by 15 July 1991 identifying the names, titles, and certificate grades of persons supervising and operating the facility, as required by Standard	

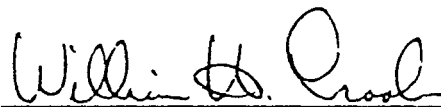
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Provision E.1, and identifying who to contact in an emergency and the appropriate means of contact.

- A. The Discharger shall submit the plan described in Standard Provision B.2 and a copy of the operation and maintenance manual for the facility by 15 July 1992.
- B. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 28 June 1991.


WILLIAM H. CROOKS, Executive Officer

JLA:cjs:6/28/91

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Composite samples shall be taken by a proportional sampling device approved by the Executive Officer or by grab samples composited in proportion to the flow. In compositing grab samples, the interval shall not exceed one hour. The Discharger may substitute 8-hour composite samples with 24-hour composite samples.

INFLUENT MONITORING

Representative samples shall be collected at approximately the same time as effluent samples. Influent monitoring shall at least include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Settleable Solids	ml/l	Grab	Daily
pH	pH units	Grab	Daily
BOD ₅ ^{1/}	mg/l	8-hour Composite	Monthly
Total Suspended Solids	mg/l	8-hour Composite	Monthly

^{1/} Five-day, 20° Celsius biochemical oxygen demand.

EFFLUENT MONITORING

Effluent samples shall be collected downstream from the final treatment unit just prior to discharge to the disposal facility or storage reservoir. Effluent samples should be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall at least include the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ^{1/}
Flow	mgd	Measured	Daily 7.0 mgd wtd
Settleable Solids	ml/l	Grab	Daily ^{min} 0.2 6.6 dwt
Dissolved Oxygen ^{2/}	mg/l	Grab	Daily 1.0 0.5 mwt
BOD ₅ ^{3/}	mg/l	8-hour Composite	2/week 40 80
Total Suspended Solids	mg/l	8-hour Composite	2/week 40 80

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<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ^{1/}
Standard Minerals ^{4/}	mg/l	Grab	Monthly
Copper	ug/l	8-hr. Composite	Monthly
Cadmium	ug/l	8-hr. Composite	Monthly
Nickel	ug/l	8-hr. Composite	Monthly
Mercury	ug/l	8-hr. Composite	Monthly
Lead	ug/l	8-hr. Composite	Monthly
Total Chromium	ug/l	8-hr. Composite	Monthly
Zinc	ug/l	8-hr. Composite	Monthly

^{1/} If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly mean for BOD), or indicate a violation and potential upset of the treatment process (e.g., less than minimum D.O.), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.

^{2/} Samples shall be collected at a depth of one foot from the storage reservoir, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

^{3/} Five-day, 20° Celsius biochemical oxygen demand.

^{4/} Standard mineral analyses shall include bicarbonate, calcium, carbonate, chloride, fluoride, iron, magnesium, nitrate, potassium, sodium, sulfate, total dissolved solids, specific conductance, pH, and total phosphorous.

SOURCE WATER MONITORING

Sampling stations shall be established within the District water supply area for derivation of a flow-weighted value for source water EC, based on the water service connections which can be related to the samples. The following shall constitute the source water monitoring program:

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<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
Specific Electrical Conductance @ 25°C ^{1/}	umhos/cm	Weekly

- ^{1/} The electrical conductance (EC) shall be reported as a flow weighted average of the EC's of all the sampling stations. Supporting calculations to determine the weighted average shall be provided with the report.

GROUND WATER MONITORING

By 15 January 1992, the Discharger shall develop a ground water monitoring network consisting of one or more background monitoring wells and three or more downgradient wells. All well locations and construction features are subject to the prior approval of the Executive Officer and must be sufficient to monitor potential impacts of the disposal operation on the uppermost ground-water aquifer. By 30 August 1991, the Discharger shall submit the proposed well design and schedule for installing the wells that will assure well development occurs on schedule.

Standards for Ground Water

Samples shall be taken monthly from approved background monitoring well(s) for one year and analyzed for the parameters specified below. After one year of sampling the wells, data from these analyses shall be reported to the Board by 15th day of the following month for use in determining water quality protection standards.

If subsequent sampling of background monitoring wells indicates significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste disposal activities, the Discharger may request modification of the water quality protection standards.

The downgradient wells shall constitute "points of compliance (POCs)". In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards. The ground water surface elevation (in feet and hundredths, M.S.L.) in all wells shall be measured on a quarterly basis and used to determine the gradient and direction of ground water flow. This information shall be displayed on a water flow net diagram for the site. Water samples shall be collected from wells in the approved monitoring network and analyzed as follows:

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<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Standard Minerals ^{1/}	mg/l	Grab	Quarterly
Heavy Metals ^{2/}	mg/l	Grab	Quarterly
VOCs ^{3/}	mg/l	Grab	Annually

^{1/} Standard mineral analyses shall include bicarbonate, calcium, carbonate, chloride, fluoride, iron, magnesium, nitrate, potassium, sodium, sulfate, total dissolved solids, specific conductance, pH, and total phosphorous.

^{2/} Heavy metal analyses shall include lead, zinc, copper, nickel, cadmium, and silver.

^{3/} Volatile Organic Compounds analyses shall include EPA's 602 compounds/constituents series.

Following each sampling event, the discharger shall determine whether there is statistically significant increase over water quality protection standards for each parameter and constituent analyzed. If the Discharger or the Board finds there is a statistically significant increase in indicator parameters or waste constituents over the water quality protection standards at the POC's, the Discharger shall notify the Board, or acknowledge the Board's findings, and submit, within 90 days, either a technical report with a plan and time schedule for implementing a verification monitoring program or a report demonstrating water quality protection standards were not exceeded. The verification Monitoring Program must be designed to verify that water quality protection standards have been exceeded and the horizontal and vertical extent of the impact.

If the Discharger, through a verification monitoring program, or the Board verifies that water quality protection standards have been exceeded at or beyond the POC's, the Discharger shall notify the Board, or acknowledge the Board's findings, and submit a technical report within 90 days. The report must contain a plan and time schedule for implementing a corrective action program designed to achieve compliance with water quality protection standards.

DISPOSAL SITE MONITORING

Disposal site monitoring shall be implemented by the Discharger no later 1 August 1991. A minimum of 20 representative locations shall be established, subject to the review and approval of the Executive Officer, for soil profile

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sampling at the reclamation area. Disposal site monitoring shall at least include the following:

<u>Measurement</u>	<u>Unit</u>	<u>Soil Profile</u>	<u>Frequency</u>
Nitrate Nitrogen	mg/kg	10 feet ^{1/}	Annually ^{2/}
Kjeldahl Nitrogen	mg/kg	10 feet ^{1/}	Annually ^{2/}
Total Salts	mg/kg	10 feet ^{1/}	Annually ^{2/}

^{1/} Samples shall be collected at approximate 1-foot increments within the root zone and at 5-foot increments thereafter to a depth of 10 feet. Sampling shall be subject to approval of the Executive Officer.

^{2/} The reclamation area shall be divided into zones which represent similar past and present uses of the property and soil conditions. Each zone shall be sampled at the end of the growing season.

The Discharger shall inspect the disposal fields weekly. During the period of 1 November through 30 April, the Discharger shall inspect the fields daily. In conducting the inspection, a log shall be kept of the condition of the area. Special attention shall be given to the preclusion of runoff outside the field in accordance with Discharge Specifications B.16. Attention shall also be given to the presence or absence of the following:

- a. Breaches of berms
- b. Vectors
- c. Erosion
- d. Standing water

A summary, addressing the items a through d listed above, shall be included with monthly monitoring reports.

SLUDGE MONITORING

When sludge is removed from ponds, but prior to disposal, a composite sample shall be analyzed, on a dry weight basis, for Total Solids (%), Nitrogen (total, $\text{NH}_4\text{-N}$, and $\text{NO}_3\text{-N}$), Total Phosphorous, Total Potassium, Total PCBs, and totals of specific metals (Pb, Zn, Cu, Ni, Cd, and Ag). Analytical results shall be submitted to the Executive Officer. Analysis of soluble concentrations of these specific metals shall also be included as needed. If final disposal is proposed to go to land, a technical report analyzing application rates and procedures relative to Department of Health Services' *Manual of Good Practices for Landspreading of Sewage Sludge* and EPA's *Process Design Manual for Land Application of Municipal Sludges* and Title 23, California Code of Regulations, Section 2511(f), shall be completed and submitted to the Executive Officer for approval.

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California Code of Regulations, Section 2511(f), shall be completed and submitted to the Executive Officer for approval.

RESERVOIR MONITORING

The freeboard shall be monitored on the storage reservoir to the nearest tenth of a foot. The following shall constitute the freeboard monitoring program.

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Frequency</u>
Freeboard	feet	Observation	Weekly

A permanent marker shall be placed in the storage reservoir with calibrations indicating the water level at design capacity and available operational freeboard.

In addition, the Discharger shall inspect the condition of the reservoir once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the reservoir surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

REPORTING

Monthly monitoring reports shall be submitted to the Regional Board by the 15th day of the following month.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during

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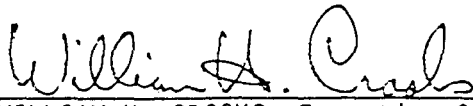
the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By 31 January of each year, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, certificate grade and general responsibilities of persons operating and maintaining the wastewater treatment plant (Standard Provision E.1).
- b. The names and telephone numbers of persons to contact regarding plant emergency and routine situations.
- c. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4).
- d. A statement identifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).
- f. A discussion of disposal area management operations, including total water application over the season; the total wastewater applied; the total nutrient loading from wastewater, sludges, and chemical fertilizers; and amount of nutrients removed through harvest of the crop. In short, a mass balance relative to pollutants of concern and hydraulic loading.

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement this monitoring program on the first day of the month following adoption of this Order.


WILLIAM H. CROOKS, Executive Officer

28 June 1991
(Date)

JA:cjs:6/28/91 AMENDED

INFORMATION SHEET

MOUNT VERNON COUNTY SANITATION DISTRICT WWTF

Mount Vernon County Sanitation District (Discharger) submitted a Report of Waste Discharge (RWD), dated 2 July 1990, and a technical report, dated June 1990, for a proposed expansion of the existing wastewater treatment and disposal facility (WWTF). The existing WWTF includes headworks, a grit chamber, two primary clarifiers, a trickling filter, a secondary clarifier, sludge digestion, and associated pumping and piping facilities. The discharge from the facility is governed by Waste Discharge Requirements (WDRs) Order No. 72-282.

WDRs Order 72-282 limits the mean daily flow to 6.6 million gallons per day (mgd). However, the existing sewage plant was originally designed to provide secondary treatment of 3.0 million gallons per day (mgd), present flows average 4.0 mgd, and influent BOD and suspended solids daily loadings have nearly doubled since 1972. The existing WWTF, therefore, does not have the organic capacity to treat all the influent waste to secondary levels, and the Discharger has been consistently in noncompliance with the 40 mg/l effluent limitations for BOD₅ and suspended solids established by WDRs Order No. 72-282 for the last 19 years. Accordingly, and along with this order, staff is recommending that the Regional Board adopt a cease and desist order requiring the Discharger to bring the discharge in compliance with effluent limits.

These requirements implement goals and guidelines prescribed by the *Water Quality Control Plan for the Tulare Lake Basin (5D)* (hereafter Basin Plan). The Basin Plan states that the minimum treatment level for WWTFs which discharge more than 1.0 mgd is secondary treatment, which is defined in the Basin Plan as 80 percent removal of BOD₅ and suspended solids, or 40 mg/l, whichever is more restrictive. The Discharger proposes to expand the WWTF to treat a design flow of 7.0 mgd and provide secondary treatment for this volume.

The Discharger proposes to treat 2.0 mgd in the existing WWTF and the remaining 5.0 mgd in the proposed improvements, which include a trickling filter and recirculation pump, two secondary clarifiers, sludge pumping facilities, an effluent structure, and associated pumping facilities as shown in Attachment B. The Discharger expects to complete the proposed project by April 1992 without interrupting the existing treatment and disposal processes during construction and completion of the improvements. The authorized flow is based upon completion of the project.

The discharge to the plant consists of domestic and industrial wastes. The industrial sources include three electroplating shops, a company which produces oilfield related chemicals, a hospital, and several auto repair shops and chemical laboratories, which are subject to categorical pretreatment standards. These requirements, therefore, require that the Discharger develop a pretreatment program.

562-1704

INFORMATION SHEET - Continued

MOUNT VERNON COUNTY
SANITATION DISTRICT WWTF
KERN COUNTY

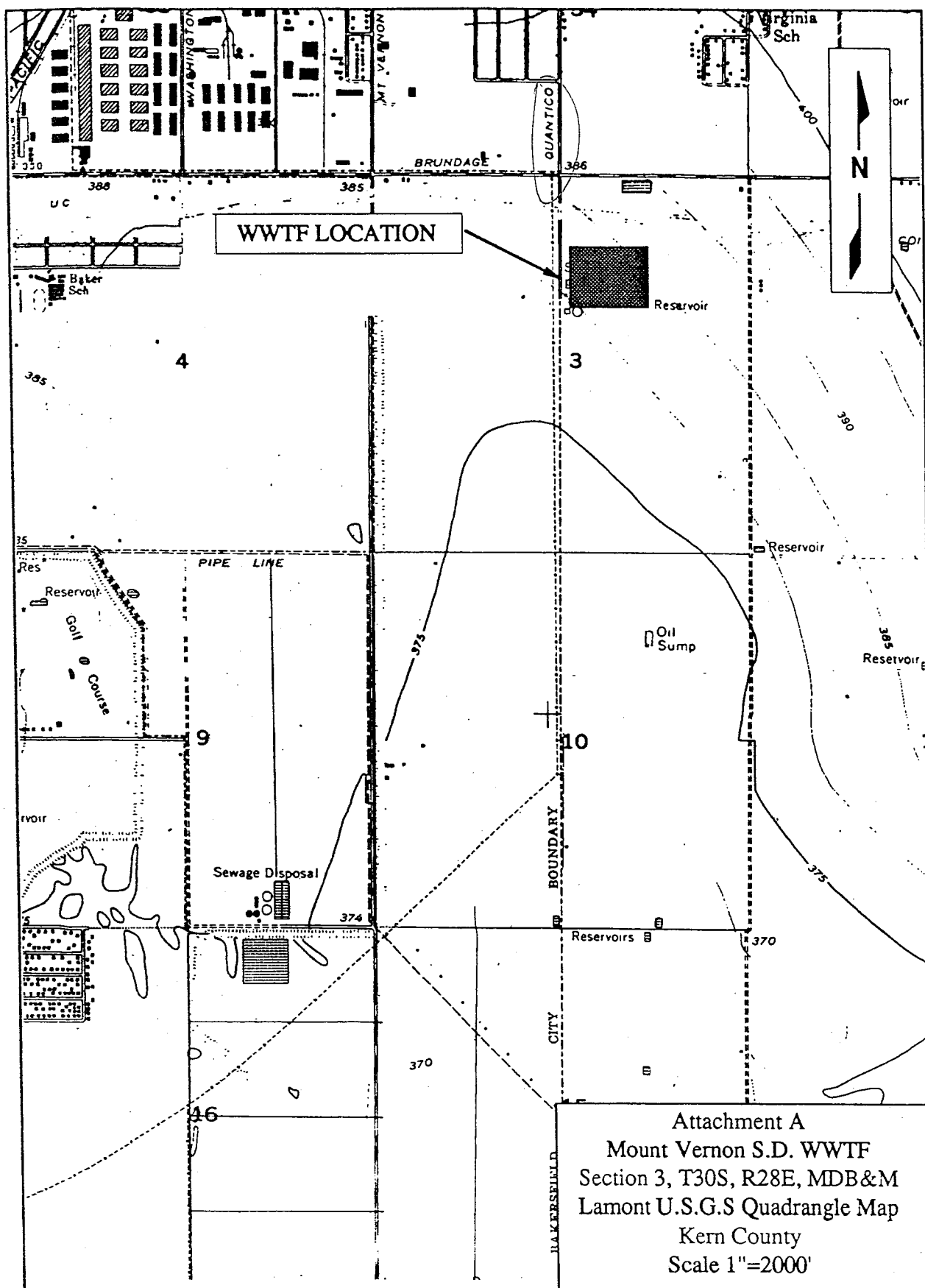
-2-

Effluent from the existing WWTF is reclaimed by surface irrigation of about 1400 acres of cotton, fiber, and seed crops on land adjacent to the WWTF and owned by the Discharger. The Discharger proposes to continue this method of wastewater reclamation, but it has not demonstrated that reclamation for the proposed 7.0-mgd discharge can be accomplished in accordance with accepted irrigation practices without contributing additional nitrates to on-site ground water. This Order requires that the Discharger submit an irrigation management plan for the use of reclaimed wastewater discharged from the WWTF. The proposed discharge for reclamation purposes is subject to reclamation criteria contained in Title 22, California Code of Regulations (CCR). These requirements prescribe reclamation requirements consistent with Title 22.

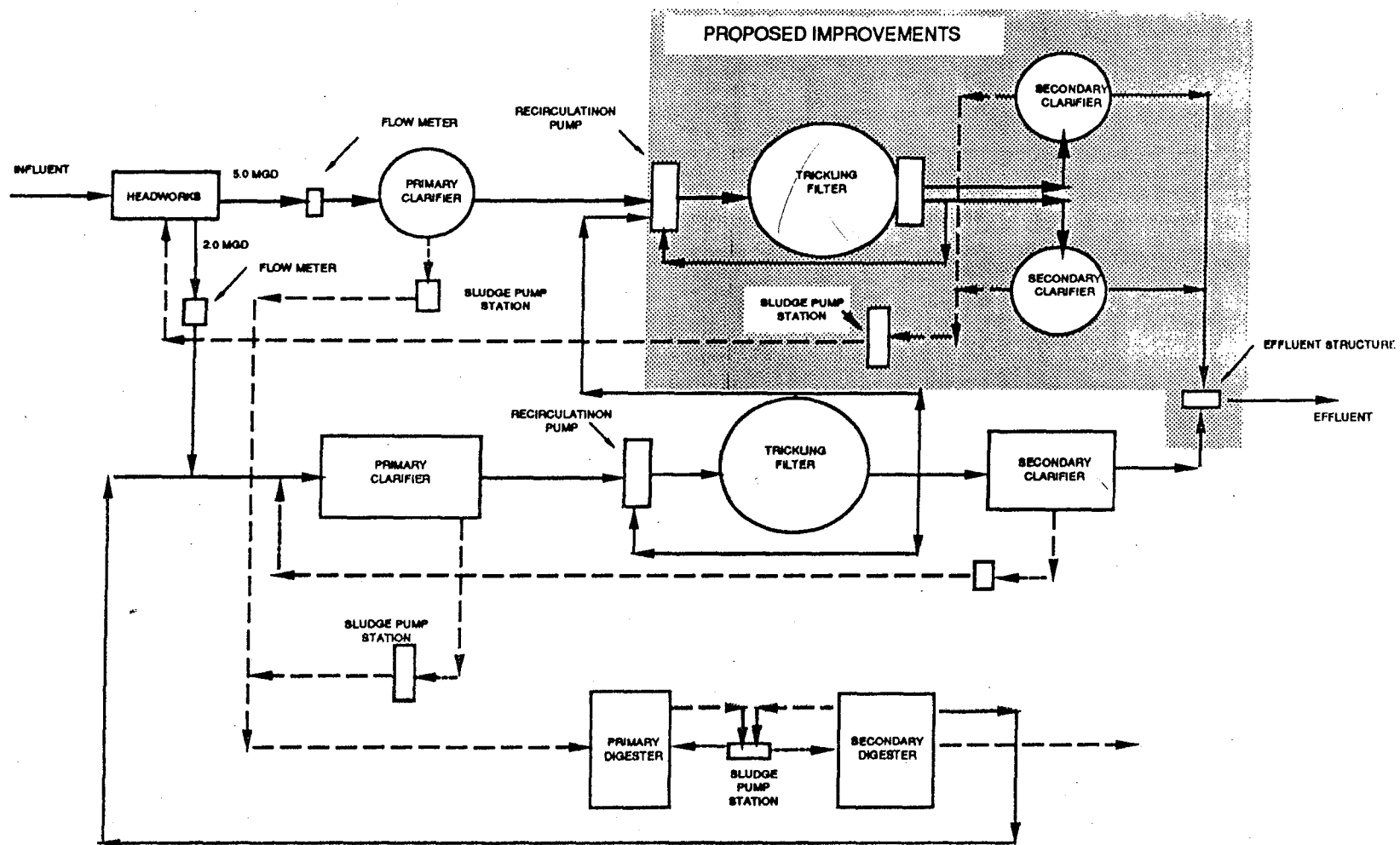
For the past 16 years, sludge from the existing plant has been disposed of by landspreading on a 20-acre parcel owned by the Discharger. Preliminary results for soil samples collected in 1986 from background samples in the sludge disposal area and from the effluent disposal area indicate that sludge from the plant contains metals in concentrations that could cause it to be classified as a "designated" waste as defined in Chapter 15, Title 23, CCR. This Order requires that the disposal of sludge comply with Chapter 15 regulations and guidelines for landspreading of sewage sludge promulgated by the State Department of Health Services (DHS). Further, this tentative Order requires that the Discharger develop and implement a ground water monitoring network to assess the effects that previous and current sludge and wastewater disposal practices have had on ground water quality.

Ground water in the area is 140 to 190 feet below the ground surface (bgs), of excellent quality (EC of about 250 umhos/cm, pH of 7.6, total dissolved solids of 160 mg/l), and moves northeast. The Basin Plan identifies beneficial uses of ground water throughout the Tulare Lake Basin as municipal and domestic supply, industrial service supply, industrial process supply, water contact recreation, and wildlife habitat. Soils below the effluent disposal area consist of sands, silty sands, silty clays, and clays, with moderate to low permeabilities.

The action to revise waste discharge requirements for the existing facility is exempt from the provisions of the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.), in accordance with Section 15301, Title 14, CCR.



SCALE: NONE



ATTACHMENT B: MOUNT VERNON COUNTY SANITATION DISTRICT WWTF IMPROVEMENTS

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD—
CENTRAL VALLEY REGION

SAN JOAQUIN WATERSHED BRANCH OFFICE:
3614 EAST ASHLAN AVENUE
FRESNO, CA 93726
PHONE: (209) 445-5116
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11 July 1991

CERTIFIED MAIL
P 400 365 878

Mr. Dale Mills
Kern County Public Works Director
Mount Vernon County Sanitation District
2700 "M" Street, Suite 500
Bakersfield, CA 93301

TRANSMITTAL OF ADOPTED WASTE DISCHARGE REQUIREMENTS AND CEASE AND DESIST ORDER
FOR MOUNT VERNON COUNTY SANITATION DISTRICT, WASTEWATER TREATMENT FACILITY, KERN
COUNTY

Enclosed is one official copy each of Order Nos. 91-155 and 91-156, as adopted
by the California Regional Water Quality Control Board, Central Valley Region,
at its 28 June 1991 meeting.

BERT E. VAN VORIS
Supervising Engineer

JLA:cjs

Enclosures: Adopted Order
Standard Provisions (Discharger only)

cc: Office of Chief Counsel, State Water Resources Control Board, Sacramento
Mr. Archie Matthews, Division of Water Quality, State Water Resources
Control Board, Sacramento
Department of Health Services, Office of Drinking Water, Fresno
Department of Fish and Game, Region IV, Fresno
Department of Water Resources, San Joaquin District, Fresno
Kern County Health Department, Bakersfield
Kern County Planning Department, Bakersfield